Business Findings & Recommendations

Retail Sales & Profitability Analysis

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Tools: MySQL, Power BI, DAX, Power Query
Dataset: Tableau Sample Superstore

1. Project Overview

The objective of this project is to analyze retail sales and profitability to identify the product categories, subcategories, and regions contributing to business performance, investigate loss-making products, and examine the relationship between discount levels and profitability.

MySQL was used to perform data validation, calculate key performance indicators, and investigate product-, discount-, and region-level performance. Power BI was used to build interactive dashboards that communicate the findings through KPI cards, charts, and a regional profitability matrix.

The analysis focuses on turning sales data into actionable business questions rather than evaluating sales volume alone.

2. Overall Business Performance

Business question: How is the business performing overall?

Evidence: The Sales Overview dashboard reports the following key performance indicators:

Metric	Result
Total Sales	**$2.33 million**
Total Profit	**$292,296.81**
Total Orders	**5,111**
Total Customers	**804**

Profit Margin	12.56%

Business implication: The business generates positive overall profit. However, the overall margin alone does not reveal which products, subcategories, or transactions may be reducing profitability.

Recommendation: Monitor sales, profit, and profit margin together. Investigate the underlying product and regional performance before making pricing or promotional decisions.

Evidence source: SQL Section 1 — Overall Business Performance; Power BI Sales Overview dashboard.

3. Discount Strategy and Profitability

Business question: Which discount levels are associated with weaker profitability?

Evidence: The SQL discount-band analysis shows that transactions with discounts above 30% generated approximately $125,507 in losses, with a profit margin of -48.22%. Transactions with discounts of 20% or less were profitable overall in the recorded analysis.

The Sales Overview dashboard includes a *Discount vs. Total Profit* chart that compares profitability across the discount levels represented in the dataset.

Business implication: Deeply discounted transactions warrant further investigation, particularly when sales are accompanied by negative profit.

Recommendation: Review highly discounted transactions at the product and subcategory levels. Investigate costs, product mix, and margins before considering changes to discount policies or promotional approval thresholds.

Evidence source: SQL Sections 5 and 9 — Profitability by Discount Band; Power BI Sales Overview dashboard.

4. Furniture Profitability

Business question: Which Furniture subcategories and regions deserve further investigation?

Evidence: The SQL analysis recorded the following results:

Furniture discount group	Total Profit	Profit Margin
Discounts of 20% or less	$74,686.28	13.38%
Discounts above 20%	-$54,956.29	-27.98%

The *Profit by Sub-Category* chart identifies Tables as a significant loss-making subcategory. The regional profitability matrix also shows that Furniture performance differs across regions, with Central Furniture recording a loss of approximately $2,802.

Business implication: Furniture profitability varies across discount groups, subcategories, and regions. The results identify high-discount Furniture transactions and Tables as priorities for further investigation.

Recommendation: Examine Tables and highly discounted Furniture transactions first. Compare product-level margins, discount levels, and regional performance before proposing changes to pricing or promotional practices.

Evidence source: SQL Sections 3, 19, 22, and 23 — Subcategory Profitability, Furniture Performance by Region, and Furniture Discount Analysis; Power BI Sales Overview and Profitability Deep Dive dashboards.

5. Regional Performance

Business question: Which regions contribute the most profit, and which have weaker margins?

Evidence: The regional SQL analysis recorded the following results:

Region	Total Profit	Profit Margin
West	$110,798.82	14.98%
East	$94,883.23	13.71%
South	$46,749.43	11.93%
Central	$39,865.31	7.92%

The Sales Overview dashboard's *Profit by Region* chart displays the same broad performance pattern.

Business implication: West contributes the highest total profit and has the strongest profit margin among the four regions. Central remains profitable overall but has a comparatively lower margin.

Recommendation: Investigate Central's category mix and product-level profitability. Compare its performance with West to identify potential areas for improvement, while accounting for differences in product mix and discount patterns.

Evidence source: SQL Sections 17 and 18 — Regional Sales and Profitability; Category Performance by Region; Power BI Sales Overview dashboard.

6. Product-Level Losses

Business question: Which individual products contribute most to total losses?

Evidence: The SQL product-level analysis ranks products with negative total profit. The results include the following examples:

Product	Total Profit	Average Discount
Cubify CubeX 3D Printer Double Head Print	-$8,879.97	53.33%
Lexmark MX611dhe Monochrome Laser Printer	-$4,689.97	40.00%
Cubify CubeX 3D Printer Triple Head Print	-$3,839.99	50.00%
Chromcraft Bull-Nose Wood Oval Conference Table	-$2,876.12	28.00%
Bush Advantage Collection Racetrack Conference Table	-$1,934.40	35.00%

The *Top 10 Loss-Making Products* visual on the Profitability Deep Dive dashboard highlights the products with the largest aggregate losses.

Business implication: Product-level analysis reveals that losses are not evenly distributed across the product range. Some products warrant closer attention because their cumulative losses are substantial.

Recommendation: Prioritize products with the largest verified losses for a detailed review of discounts, sales, transaction-level profit, and available cost information. Investigate whether losses are concentrated in specific transactions before deciding whether pricing, promotional practices, or other factors require changes.

Evidence source: SQL Sections 10–16 — Loss-Making Products and Product-Level Discount Analysis; Power BI Profitability Deep Dive dashboard.

7. Dashboard Overview

The project contains two Power BI report pages.

Sales Overview

- **KPI cards for total sales, profit, orders, customers, and profit margin.**
- **Monthly sales by calendar month, aggregated across the years in the dataset.**
- **Profit by product category.**
- **Discount versus total profit.**
- **Regional category profitability matrix.**
- **Profit by region.**

Profitability Deep Dive

- **Profit by subcategory, distinguishing profitable and loss-making subcategories.**
- **Top 10 loss-making products based on aggregate product profit.**
- **Profit margin by subcategory.**

Together, these pages provide a high-level view of business performance and a more detailed investigation of profitability risks.

8. Limitations

This analysis uses the Tableau Sample Superstore dataset. The results are illustrative and should not be interpreted as an audit of a real company's current financial performance.

The analysis identifies associations between discount levels and profitability but does not establish that discounts alone caused the losses. Product costs, product mix, transaction characteristics, and regional differences may also affect profitability.

The monthly sales chart aggregates records by calendar month across years. It is therefore useful for comparing monthly patterns but should not be interpreted as a chronological year-by-year trend.

The recommendations are proposed areas for investigation. Their potential financial impact should be evaluated with further analysis or testing before implementation.

9. Conclusion

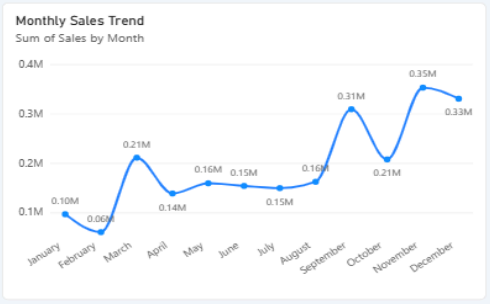
This project demonstrates how SQL and Power BI can be combined to analyze retail performance and communicate business findings.

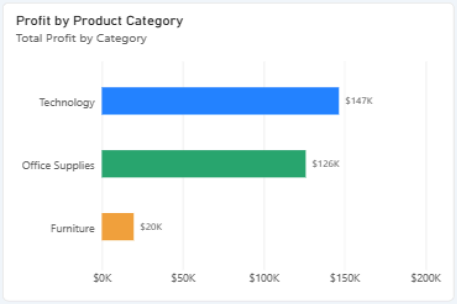
The analysis identifies overall sales and profitability, regional differences, weaker Furniture performance, and product-level losses. It also highlights an association between higher discount levels and weaker profitability in parts of the dataset.

The findings provide a basis for further investigation into product profitability, discounting practices, and regional performance. Any operational changes should be based on validated data and tested before implementation.

Retail Sales & Profitability Analysis

Total Sales	Total Profit	Total Orders1	Total Customers	Profit Margin
$2.33M	$292.30K	5,111	804	12.56%

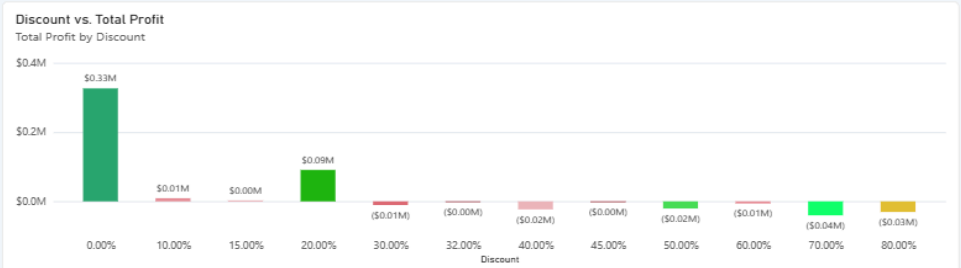


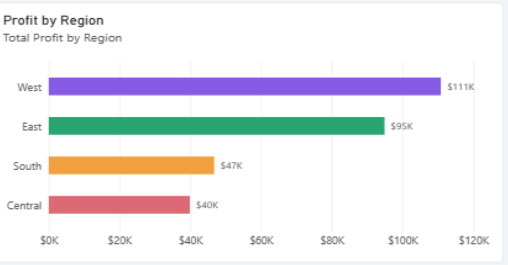


Categories

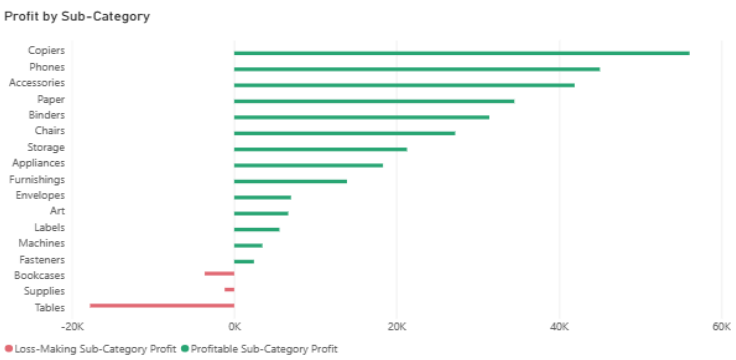
- ☐ **Furniture**
- ☐ **Office Supplies**
- ☐ **Technology**

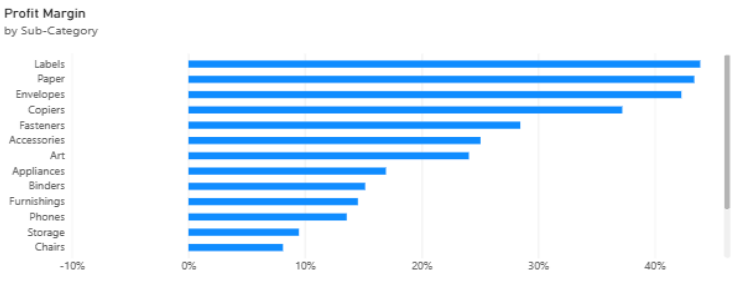
Region	Furniture	Office Supplies	Technology	**Total**
Central	($2,802.21)	$8,970.08	$33,697.43	**$39,865.31**
East	$3,444.74	$42,996.74	$48,441.78	**$94,883.26**
South	$6,771.21	$19,986.39	$19,991.83	**$46,749.43**
West	$12,316.25	$54,070.23	$44,412.34	**$1,10,798.82**
Total	**$19,730.00**	**$1,26,023.44**	**$1,46,543.38**	**$2,92,296.81**

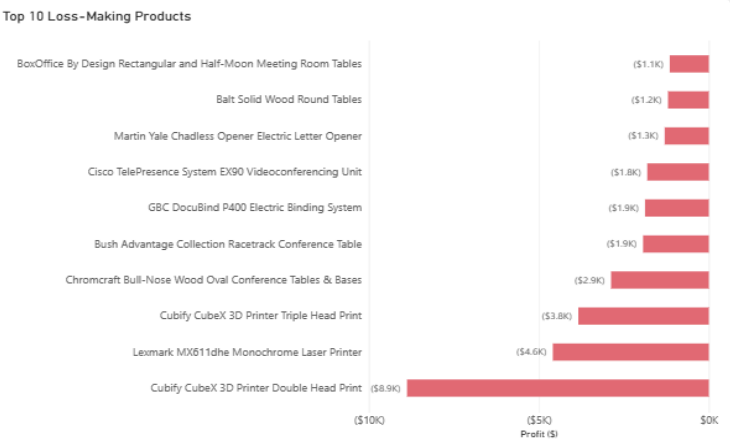




Profitability Deep Dive







Product Name	Sum of Profit	Total Profit
Cubify CubeX 3D Printer Double Head Print	-8,879.97	($8,879.97)
Total	**-8,879.97**	**($8,879.97)**